

35. $15 \times (4 - 1)^2 + 5 \times (17 + 2 \times 3) + 4$
36. $18 \times (12 - 6)^4 - 8 \times (30 - 14 \times 16) - 90$
37. $4 \times (5 + 2 \times (6 + 8)) + 10 \times 3^2$
38. $9 \times (13 - 12 \times (41 - 32)) - 8 \times 45$
39. $21 - 3 \times (5 \times (2 + 6) - 3 \times (18 - 11) + 25) \div 2$
40. $48 - 6 \times (10 \times (18 + 12) - 25 \times (16 - 9) + 19) \div 4$

3.4 Rational Numbers

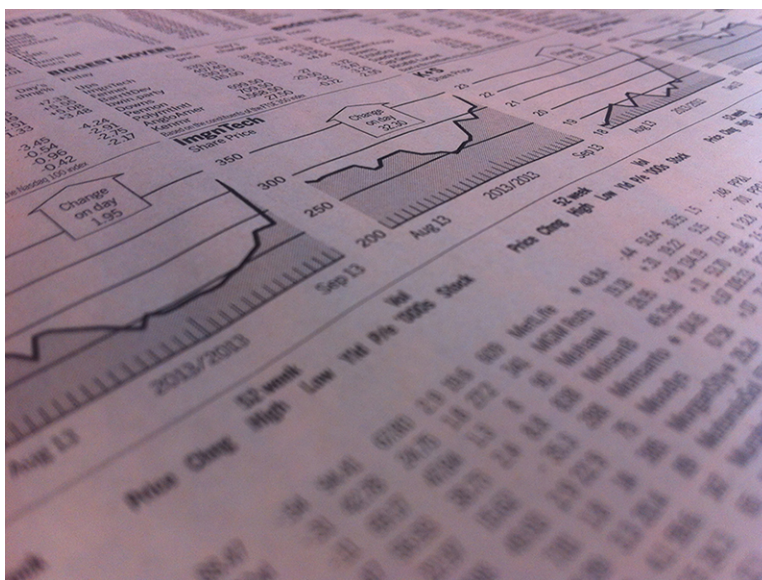


Figure 3.20 Stock gains and losses are often represented as percentages.(credit: "stock market quotes in newspaper" by Andreas Poike/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Define and identify numbers that are rational.
2. Simplify rational numbers and express in lowest terms.
3. Add and subtract rational numbers.
4. Convert between improper fractions and mixed numbers.
5. Convert rational numbers between decimal and fraction form.
6. Multiply and divide rational numbers.
7. Apply the order of operations to rational numbers to simplify expressions.
8. Apply density property of rational numbers.
9. Solve problems involving rational numbers.
10. Use fractions to convert between units.
11. Define and apply percent.
12. Solve problems using percent.

We are often presented with percentages or fractions to explain how much of a population has a certain feature. For example, the 6-year graduation rate of college students at public institutions is 57.6%, or $\frac{72}{125}$. That fraction may be unsettling. But without the context, the percentage is hard to judge. So how does that compare to private institutions? There, the 6-year graduation rate is 65.4%, or $\frac{327}{500}$. Comparing the percentages is straightforward, but the fractions are harder to interpret due to different denominators. For more context, historical data could be found. One study reported that the 6-year graduation rate in 1995 was 56.4%. Comparing that historical number to the recent 6-year graduation rate at public institutions of 57.6% shows that there hasn't been much change in that rate.

Defining and Identifying Numbers That Are Rational

A **rational number** (called rational since it is a ratio) is just a fraction where the numerator is an integer and the denominator is a non-zero integer. As simple as that is, they can be represented in many ways. It should be noted here that any integer is a rational number. An integer, n , written as a fraction of two integers is $\frac{n}{1}$.

In its most basic representation, a rational number is an integer divided by a non-zero integer, such as $\frac{3}{12}$. Fractions may

be used to represent parts of a whole. The denominator is the total number of parts to the object, and the numerator is how many of those parts are being used or selected. So, if a pizza is cut into 8 equal pieces, each piece is $\frac{1}{8}$ of the pizza. If you take three slices, you have $\frac{3}{8}$ of the pizza (Figure 3.21). Similarly, if in a group of 20 people, 5 are wearing hats, then $\frac{5}{20}$ of the people are wearing hats (Figure 3.22).

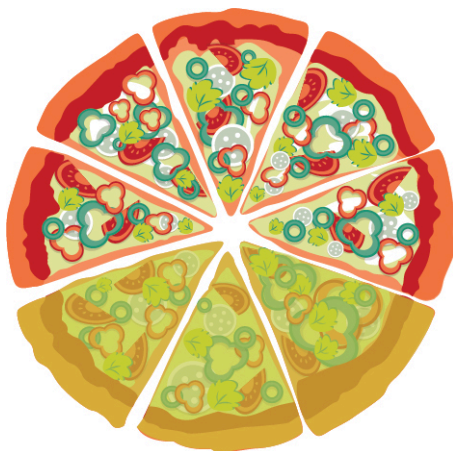


Figure 3.21 Pizza cut in 8 slices, with 3 slices highlighted



Figure 3.22 Group of 20 people, with 5 people wearing hats

Another representation of rational numbers is as a mixed number, such as $2\frac{5}{8}$ (Figure 3.23). This represents a whole number (2 in this case), plus a fraction (the $\frac{5}{8}$).



Figure 3.23 Two whole pizzas and one partial pizza

Rational numbers may also be expressed in decimal form; for instance, as 1.34. When 1.34 is written, the decimal part, 0.34, represents the fraction $\frac{34}{100}$, and the number 1.34 is equal to $1\frac{34}{100}$. However, not all decimal representations are rational numbers.

A number written in decimal form where there is a last decimal digit (after a given decimal digit, all following decimal digits are 0) is a **terminating decimal**, as in 1.34 above. Alternately, any decimal numeral that, after a finite number of decimal digits, has digits equal to 0 for all digits following the last non-zero digit.

All numbers that can be expressed as a terminating decimal are rational. This comes from what the decimal represents. The decimal part is the fraction of the decimal part divided by the appropriate power of 10. That power of 10 is the number of decimal digits present, as for 0.34, with two decimal digits, being equal to $\frac{34}{100}$.

Another form that is a rational number is a decimal that repeats a pattern, such as 67.1313... When a rational number is expressed in decimal form and the decimal is a repeated pattern, we use special notation to designate the part that repeats. For example, if we have the repeating decimal 4.3636..., we write this as $4.\overline{36}$. The bar over the 36 indicates that the 36 repeats forever.

If the decimal representation of a number does not terminate or form a repeating decimal, that number is not a rational number.

One class of numbers that is not rational is the **square roots** of integers or rational numbers that are not **perfect squares**, such as $\sqrt{10}$ and $\sqrt{\frac{25}{6}}$. More generally, the number b is the square root of the number a if $a = b^2$. The notation for this is $b = \sqrt{a}$, where the symbol $\sqrt{}$ is the square root sign. An integer perfect square is any integer that can be written as the square of another integer. A rational perfect square is any rational number that can be written as a fraction of two integers that are perfect squares.

Sometimes you may be able to identify a perfect square from memory. Another process that may be used is to factor the number into the product of an integer with itself. Or a calculator (such as Desmos) may be used to find the square root of the number. If the calculator yields an integer, the original number was a perfect square.

TECH CHECK

Using Desmos to Find the Square Root of a Number

When Desmos is used, there is a tab at the bottom of the screen that opens the keyboard for Desmos. The keyboard is shown below. On the keyboard ([Figure 3.24](#)) is the square root symbol ($\sqrt{}$). To find the square root of a number, click the square root key, and then type the number. Desmos will automatically display the value of the square root as you enter the number.

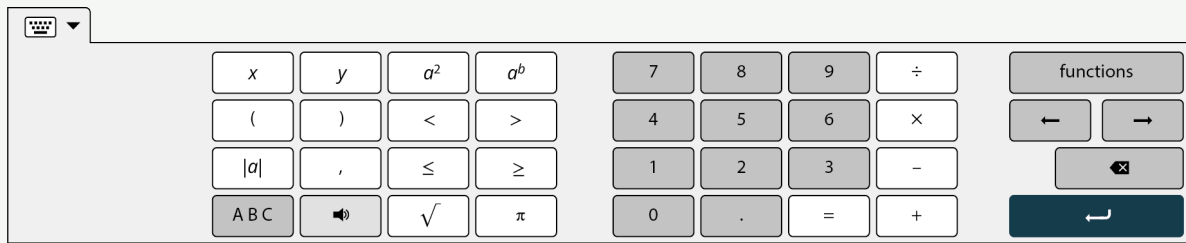


Figure 3.24 Desmos keyboard with square root key

EXAMPLE 3.51**Identifying Perfect Squares**

Which of the following are perfect squares?

1. 45
2. 144

✓ **Solution**

1. We could attempt to find the perfect square by factoring. Writing all the factor pairs of 45 results in 1×45 , 3×15 , and 5×9 . None of the pairs is a square, so 45 is not a perfect square. Using a calculator to find the square root of 45, we obtain 6.708 (rounded to three decimal places). Since this was not an integer, the original number was not a perfect square.
2. We could attempt to find the perfect square by factoring. Writing all the factor pairs of 144 results in 1×144 , 2×72 , 3×48 , 6×24 , 8×18 , and 12×12 . Since the last pair is an integer multiplied by itself, 144 is a perfect square. Alternately, using Desmos to find the square root of 144, we obtain 12. Since the square root of 144 is an integer, 144 is a perfect square.

YOUR TURN 3.51

Determine if the following are perfect squares:

1. 94
2. 441

VIDEO

[Introduction to Fractions \(https://openstax.org/r/Introduction_to_Fractions\)](https://openstax.org/r/Introduction_to_Fractions)

EXAMPLE 3.52**Identifying Rational Numbers**

Determine which of the following are rational numbers:

1. $\sqrt{73}$
2. 4.556
3. $3\frac{1}{5}$
4. $\frac{41}{17}$
5. $5.\overline{64}$

✓ **Solution**

1. Since 73 is not a perfect square, its square root is not a rational number. This can also be seen when a calculator is used. Entering $\sqrt{73}$ into a calculator results in 8.544003745317 (and then more decimal values after that). There is no repeated pattern, so this is not a rational number.
2. Since 4.556 is a decimal that terminates, this is a rational number.

3. $3\frac{1}{5}$ is a mixed number, so it is a rational number.
4. $\frac{41}{17}$ is an integer divided by an integer, so it is a rational number.
5. 5.646464... is a decimal that repeats a pattern, so it is a rational number.

> YOUR TURN 3.52

Determine which of the following are rational numbers:

1. $\sqrt{13}$
2. $-13.\overline{21}$
3. $-\frac{48}{16}$
4. $-4\frac{18}{19}$
5. 14.1131

Simplifying Rational Numbers and Expressing in Lowest Terms

A rational number is one way to express the division of two integers. As such, there may be multiple ways to express the same value with different rational numbers. For instance, $\frac{4}{5}$ and $\frac{12}{15}$ are the same value. If we enter them into a calculator, they both equal 0.8. Another way to understand this is to consider what it looks like in a figure when two fractions are equal.

In [Figure 3.25](#), we see that $\frac{3}{5}$ of the rectangle and $\frac{9}{15}$ of the rectangle are equal areas.

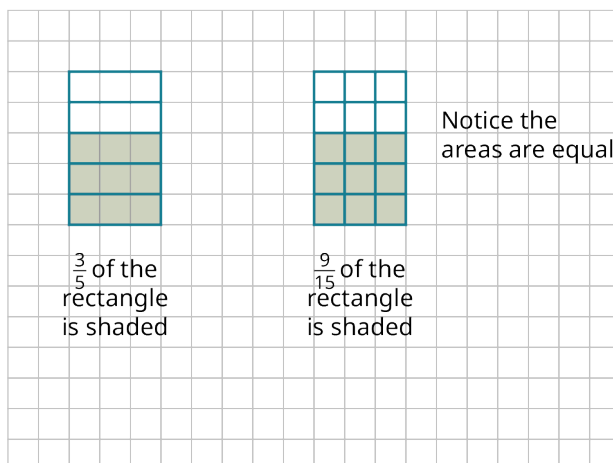


Figure 3.25 Two Rectangles with Equal Areas

They are the same proportion of the area of the rectangle. The left rectangle has 5 pieces, three of which are shaded. The right rectangle has 15 pieces, 9 of which are shaded. Each of the pieces of the left rectangle was divided equally into three pieces. This was a multiplication. The numerator describing the left rectangle was 3 but it becomes 3×3 , or 9, as each piece was divided into three. Similarly, the denominator describing the left rectangle was 5, but became 5×3 , or 15, as each piece was divided into 3. The fractions $\frac{3}{5}$ and $\frac{9}{15}$ are **equivalent** because they represent the same portion (often loosely referred to as equal).

This understanding of equivalent fractions is very useful for conceptualization, but it isn't practical, in general, for determining when two fractions are equivalent. Generally, to determine if the two fractions $\frac{a}{b}$ and $\frac{c}{d}$ are equivalent, we check to see that $a \times d = b \times c$. If those two products are equal, then the fractions are equal also.

EXAMPLE 3.53

Determining If Two Fractions Are Equivalent

Determine if $\frac{12}{30}$ and $\frac{14}{35}$ are equivalent fractions.

 Solution

Applying the definition, $a = 12$, $b = 30$, $c = 14$ and $d = 35$. So $a \times d = 12 \times 35 = 420$. Also, $b \times c = 30 \times 14 = 420$. Since these values are equal, the fractions are equivalent.

 YOUR TURN 3.53

1. Determine if $\frac{8}{14}$ and $\frac{12}{26}$ are equivalent fractions.

That $a \times d = b \times c$ indicates the fractions $\frac{a}{b}$ and $\frac{c}{d}$ are equivalent is due to some algebra. One property of natural numbers, integers, and rational numbers (also irrational numbers) is that for any three numbers a , b , and c with $c \neq 0$, if $a = b$, then $a/c = b/c$. In other words, when two numbers are equal, then dividing both numbers by the same non-zero number, the two newly obtained numbers are also equal. We can apply that to $a \times d$ and $b \times c$, to show that $\frac{a}{b}$ and $\frac{c}{d}$ are equivalent if $a \times d = b \times c$.

If $a \times d = b \times c$, and $c \neq 0$, $d \neq 0$, we can divide both sides by and obtain the following: $\frac{a \times d}{c} = \frac{b \times c}{c}$. We can divide out the c on the right-hand side of the equation, resulting in $\frac{a \times d}{c} = b$. Similarly, we can divide both sides of the equation by d and obtain the following: $\frac{a \times d}{c \times d} = \frac{b}{d}$. We can divide out d on the left-hand side of the equation, resulting in $\frac{a}{c} = \frac{b}{d}$. So, the rational numbers $\frac{a}{c}$ and $\frac{b}{d}$ are equivalent when $a \times d = b \times c$.

 VIDEO

[Equivalent Fractions \(https://openstax.org/r/Equivalent_Fractions\)](https://openstax.org/r/Equivalent_Fractions)

Recall that a **common divisor** or **common factor** of a set of integers is one that divides all the numbers of the set of numbers being considered. In a fraction, when the numerator and denominator have a common divisor, that common divisor can be **divided out**. This is often called **canceling the common factors** or, more colloquially, as **canceling**.

To show this, consider the fraction $\frac{36}{63}$. The numerator and denominator have the common factor 3. We can rewrite the fraction as $\frac{36}{63} = \frac{12 \times 3}{21 \times 3}$. The common divisor 3 is then divided out, or canceled, and we can write the fraction as

$\frac{12 \times \cancel{3}}{21 \times \cancel{3}} = \frac{12}{21}$. The 3s have been crossed out to indicate they have been divided out. The process of dividing out two

factors is also referred to as **reducing the fraction**.

If the numerator and denominator have no common positive divisors other than 1, then the rational number is in **lowest terms**.

The process of dividing out common divisors of the numerator and denominator of a fraction is called **reducing the fraction**.

One way to reduce a fraction to lowest terms is to determine the GCD of the numerator and denominator and divide out the GCD. Another way is to divide out common divisors until the numerator and denominator have no more common factors.

EXAMPLE 3.54**Reducing Fractions to Lowest Terms**

Express the following rational numbers in lowest terms:

1. $\frac{36}{48}$
2. $\frac{100}{250}$
3. $\frac{51}{136}$

 Solution

1. One process to reduce $\frac{36}{48}$ to lowest terms is to identify the GCD of 36 and 48 and divide out the GCD. The GCD of 36 and 48 is 12.

Step 1: We can then rewrite the numerator and denominator by factoring 12 from both.

$$\frac{36}{48} = \frac{12 \times 3}{12 \times 4}$$

Step 2: We can now divide out the 12s from the numerator and denominator.

$$\frac{36}{48} = \frac{\cancel{12} \times 3}{\cancel{12} \times 4} = \frac{3}{4}$$

So, when $\frac{36}{48}$ is reduced to lowest terms, the result is $\frac{3}{4}$.

Alternately, you could identify a common factor, divide out that common factor, and repeat the process until the remaining fraction is in lowest terms.

Step 1: You may notice that 4 is a common factor of 36 and 48.

Step 2: Divide out the 4, as in $\frac{36}{48} = \frac{4 \times 9}{4 \times 12} = \frac{\cancel{4} \times 9}{\cancel{4} \times 12} = \frac{9}{12}$.

Step 3: Examining the 9 and 12, you identify 3 as a common factor and divide out the 3, as in $\frac{9}{12} = \frac{\cancel{3} \times 3}{\cancel{3} \times 4} = \frac{3}{4}$. The 3 and 4 have no common positive factors other than 1, so it is in lowest terms.

So, when $\frac{36}{48}$ is reduced to lowest terms, the result is $\frac{3}{4}$.

2. **Step 1:** To reduce $\frac{100}{250}$ to lowest terms, identify the GCD of 100 and 250. This GCD is 50.

Step 2: We can then rewrite the numerator and denominator by factoring 50 from both.

$$\frac{100}{250} = \frac{50 \times 2}{50 \times 5}$$

Step 3: We can now divide out the 50s from the numerator and denominator.

$$\frac{100}{250} = \frac{\cancel{50} \times 2}{\cancel{50} \times 5} = \frac{2}{5}$$

So, when $\frac{100}{250}$ is reduced to lowest terms, the result is $\frac{2}{5}$.

3. **Step 1:** To reduce $\frac{51}{136}$ to lowest terms, identify the GCD of 51 and 136. This GCD is 17.

Step 2: We can then rewrite the numerator and denominator by factoring 17 from both.

$$\frac{51}{136} = \frac{17 \times 3}{17 \times 8}$$

Step 3: We can now divide out the 17s from the numerator and denominator.

$$\frac{51}{136} = \frac{\cancel{17} \times 3}{\cancel{17} \times 8} = \frac{3}{8}$$

So, when $\frac{51}{136}$ is reduced to lowest terms, the result is $\frac{3}{8}$.

YOUR TURN 3.54

1. Express $\frac{252}{840}$ in lowest terms.

VIDEO

Reducing Fractions to Lowest Terms (https://openstax.org/r/Reducing_Fractions_to_Lowest_Terms)

TECH CHECK

Using Desmos to Find Lowest Terms

Desmos is a free [online calculator](https://openstax.org/r/calculator) (<https://openstax.org/r/calculator>). Desmos supports reducing fractions to lowest terms. When a fraction is entered, Desmos immediately calculates the decimal representation of the fraction.

However, to the left of the fraction, there is a button that, when clicked, shows the fraction in reduced form.

 VIDEO

[Using Desmos to Reduce a Fraction \(https://openstax.org/r/Using_Desmos_to_Reduce_a_Fraction\)](https://openstax.org/r/Using_Desmos_to_Reduce_a_Fraction)

Adding and Subtracting Rational Numbers

Adding or subtracting rational numbers can be done with a calculator, which often returns a decimal representation, or by finding a common denominator for the rational numbers being added or subtracted.

 TECH CHECK

Using Desmos to Add Rational Numbers in Fractional Form

To create a fraction in Desmos, enter the numerator, then use the division key (/) on your keyboard, and then enter the denominator. The fraction is then entered. Then click the right arrow key to exit the denominator of the fraction. Next, enter the arithmetic operation (+ or -). Then enter the next fraction. The answer is displayed dynamically (calculates as you enter). To change the Desmos result from decimal form to fractional form, use the fraction button ([Figure 3.26](#)) on the left of the line that contains the calculation:



Figure 3.26 Fraction button on the Desmos keyboard

EXAMPLE 3.55

Adding Rational Numbers Using Desmos

Calculate $\frac{23}{42} + \frac{9}{56}$ using Desmos.

 Solution

Enter $\frac{23}{42} + \frac{9}{56}$ in Desmos. The result is displayed as 0.7083333333 (which is $0.708\overline{3}$). Clicking the fraction button to the left on the calculation line yields $\frac{17}{24}$.

 YOUR TURN 3.55

1. Calculate $\frac{124}{297} + \frac{3}{125}$.

Performing addition and subtraction without a calculator may be more involved. When the two rational numbers have a common denominator, then adding or subtracting the two numbers is straightforward. Add or subtract the numerators, and then place that value in the numerator and the common denominator in the denominator. Symbolically, we write this as $\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}$. This can be seen in the [Figure 3.27](#), which shows $\frac{3}{20} + \frac{4}{20} = \frac{7}{20}$.

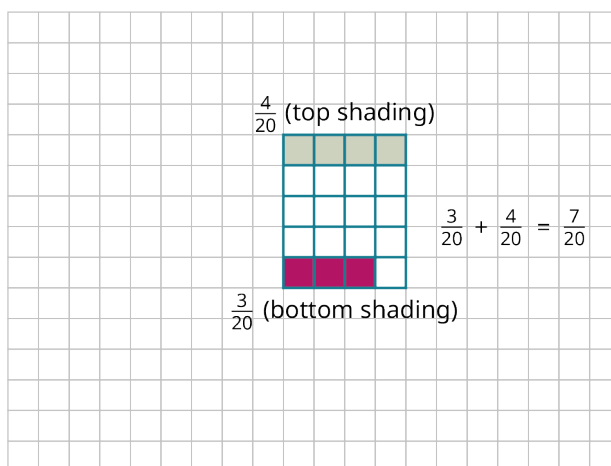


Figure 3.27 Partially Shaded Rectangle

It is customary to then write the result in lowest terms.

FORMULA

If c is a non-zero integer, then $\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}$.

EXAMPLE 3.56**Adding Rational Numbers with the Same Denominator**

Calculate $\frac{13}{28} + \frac{7}{28}$.

✓ **Solution**

Since the rational numbers have the same denominator, we perform the addition of the numerators, $13 + 7$, and then place the result in the numerator and the common denominator, 28, in the denominator. $\frac{13}{28} + \frac{7}{28} = \frac{13+7}{28} = \frac{20}{28}$

Once we have that result, reduce to lowest terms, which gives $\frac{20}{28} = \frac{4 \times 5}{4 \times 7} = \frac{\cancel{4} \times 5}{\cancel{4} \times 7} = \frac{5}{7}$.

>

YOUR TURN 3.56

1. Calculate $\frac{38}{73} + \frac{7}{73}$.

EXAMPLE 3.57**Subtracting Rational Numbers with the Same Denominator**

Calculate $\frac{45}{136} - \frac{17}{136}$.

✓ **Solution**

Since the rational numbers have the same denominator, we perform the subtraction of the numerators, $45 - 17$, and then place the result in the numerator and the common denominator, 136, in the denominator.

$$\frac{45}{136} - \frac{17}{136} = \frac{45-17}{136} = \frac{28}{136}$$

Once we have that result, reduce to lowest terms, this gives $\frac{28}{136} = \frac{4 \times 7}{4 \times 34} = \frac{\cancel{4} \times 7}{\cancel{4} \times 34} = \frac{7}{34}$.

> YOUR TURN 3.57

1. Calculate $\frac{21}{40} - \frac{8}{40}$.

When the rational numbers do not have common denominators, then we have to transform the rational numbers so that they do have common denominators. The common denominator that reduces work later in the problem is the LCM of the numerator and denominator. When adding or subtracting the rational numbers $\frac{a}{b}$ and $\frac{c}{d}$, we perform the following steps.

Step 1: Find $LCM(b, d)$.

Step 2: Calculate $n = \frac{LCM(b,d)}{b}$ and $m = \frac{LCM(b,d)}{d}$.

Step 3: Multiply the numerator and denominator of $\frac{a}{b}$ by n , yielding $\frac{a \times n}{b \times n}$.

Step 4: Multiply the numerator and denominator of $\frac{c}{d}$ by m , yielding $\frac{c \times m}{d \times m}$.

Step 5: Add or subtract the rational numbers from Steps 3 and 4, since they now have the common denominators.

You should be aware that the common denominator is $LCM(b, d)$. For the first denominator, we have $b \times n = b \times \frac{LCM(b,d)}{b} = LCM(b, d)$, since we multiply and divide $LCM(b, d)$ by the same number. For the same reason, $d \times m = d \times \frac{LCM(b,d)}{d} = LCM(b, d)$.

EXAMPLE 3.58**Adding Rational Numbers with Unequal Denominators**

Calculate $\frac{11}{18} + \frac{2}{15}$.

✓ Solution

The denominators of the fractions are 18 and 15, so we label $b = 18$ and $d = 15$.

Step 1: Find $LCM(18, 15)$. This is 90.

Step 2: Calculate n and m . $n = \frac{90}{18} = 5$ and $m = \frac{90}{15} = 6$.

Step 3: Multiplying the numerator and denominator of $\frac{11}{18}$ by $n = 5$ yields $\frac{11 \times 5}{18 \times 5} = \frac{55}{90}$.

Step 4: Multiply the numerator and denominator of $\frac{2}{15}$ by $m = 6$ yields $\frac{2 \times 6}{15 \times 6} = \frac{12}{90}$.

Step 5: Now we add the values from Steps 3 and 4: $\frac{55}{90} + \frac{12}{90} = \frac{67}{90}$.

This is in lowest terms, so we have found that $\frac{11}{18} + \frac{2}{15} = \frac{67}{90}$.

> YOUR TURN 3.58

1. Calculate $\frac{4}{9} + \frac{7}{12}$.

EXAMPLE 3.59**Subtracting Rational Numbers with Unequal Denominators**

Calculate $\frac{14}{25} - \frac{9}{70}$.

✓ Solution

The denominators of the fractions are 25 and 70, so we label $b = 25$ and $d = 70$.

Step 1: Find $LCM(25, 70)$. This is 350.

Step 2: Calculate n and m : $n = \frac{350}{25} = 14$ and $m = \frac{350}{70} = 5$.

Step 3: Multiplying the numerator and denominator of $\frac{14}{25}$ by $n = 14$ yields $\frac{14 \times 14}{25 \times 14} = \frac{196}{350}$.

Step 4: Multiplying the numerator and denominator of $\frac{9}{70}$ by $m = 5$ yields $\frac{9 \times 5}{70 \times 5} = \frac{45}{350}$.

Step 5: Now we subtract the value from Step 4 from the value in Step 3: $\frac{196}{350} - \frac{45}{350} = \frac{151}{350}$.

This is in lowest terms, so we have found that $\frac{14}{25} - \frac{9}{70} = \frac{151}{350}$.

YOUR TURN 3.59

1. Calculate $\frac{10}{99} - \frac{17}{300}$.

VIDEO

[Adding and Subtracting Fractions with Different Denominators \(https://openstax.org/r/Adding_and_Subtracting_Fractions\)](https://openstax.org/r/Adding_and_Subtracting_Fractions)

Converting Between Improper Fractions and Mixed Numbers

One way to visualize a fraction is as parts of a whole, as in $\frac{5}{12}$ of a pizza. But when the numerator is larger than the denominator, as in $\frac{23}{12}$, then the idea of parts of a whole seems not to make sense. Such a fraction is an **improper fraction**. That kind of fraction could be written as an integer plus a fraction, which is a **mixed number**. The fraction $\frac{23}{12}$ rewritten as a mixed number would be $1\frac{11}{12}$. Arithmetically, $1\frac{11}{12}$ is equivalent to $1 + \frac{11}{12}$, which is read as “one and 11 twelfths.”

Improper fractions can be rewritten as mixed numbers using division and remainders. To find the mixed number representation of an improper fraction, divide the numerator by the denominator. The quotient is the integer part, and the remainder becomes the numerator of the remaining fraction.

EXAMPLE 3.60

Rewriting an Improper Fraction as a Mixed Number

Rewrite $\frac{48}{13}$ as a mixed number.

Solution

When 48 is divided by 13, the result is 3 with a remainder of 9. So, we can rewrite $\frac{48}{13}$ as $3\frac{9}{13}$.

YOUR TURN 3.60

1. Rewrite $\frac{95}{26}$ as a mixed number.

VIDEO

[Converting an Improper Fraction to a Mixed Number Using Desmos \(https://openstax.org/r/Converting_an_Improper_Fraction_to_a_Mixed_Number_Using_Desmos\)](https://openstax.org/r/Converting_an_Improper_Fraction_to_a_Mixed_Number_Using_Desmos)

Similarly, we can convert a mixed number into an improper fraction. To do so, first convert the whole number part to a fraction by writing the whole number as itself divided by 1, and then add the two fractions.

Alternately, we can multiply the whole number part and the denominator of the fractional part. Next, add that product to the numerator. Finally, express the number as that product divided by the denominator.

EXAMPLE 3.61**Rewriting a Mixed Number as an Improper Fraction**

Rewrite $5\frac{4}{9}$ as an improper fraction.

 Solution

Step 1: Multiply the integer part, 5, by the denominator, 9, which gives $5 \times 9 = 45$.

Step 2: Add that product to the numerator, which gives $45 + 4 = 49$.

Step 3: Write the number as the sum, 49, divided by the denominator, 9, which gives $\frac{49}{9}$.

 YOUR TURN 3.61

1. Rewrite $9\frac{5}{14}$ as an improper fraction.

 TECH CHECK**Using Desmos to Rewrite a Mixed Number as an Improper Fraction**

Desmos can be used to convert from a mixed number to an improper fraction. To do so, we use the idea that a mixed number, such as $5\frac{6}{11}$, is another way to represent $5 + \frac{6}{11}$. If $5 + \frac{6}{11}$ is entered in Desmos, the result is the decimal form of the number. However, clicking the fraction button to the left will convert the decimal to an improper fraction, $\frac{61}{11}$. As an added bonus, Desmos will automatically reduce the fraction to lowest terms.

Converting Rational Numbers Between Decimal and Fraction Forms

Understanding what decimals represent is needed before addressing conversions between the fractional form of a number and its **decimal form**, or writing a number in **decimal notation**. The decimal number 4.557 is equal to $4\frac{557}{1,000}$. The decimal portion, .557, is 557 divided by 1,000. To write any decimal portion of a number expressed as a terminating decimal, divide the decimal number by 10 raised to the power equal to the number of decimal digits. Since there were three decimal digits in 4.557, we divided 557 by $10^3 = 1000$.

Decimal representations may be very long. It is convenient to **round off** the decimal form of the number to a certain number of decimal digits. To round off the decimal form of a number to n (decimal) digits, examine the $(n + 1)$ st decimal digit. If that digit is 0, 1, 2, 3, or 4, the number is rounded off by writing the number to the n th decimal digit and no further. If the $(n + 1)$ st decimal digit is 5, 6, 7, 8, or 9, the number is rounded off by writing the number to the n th digit, then replacing the n th digit by one more than the n th digit.

EXAMPLE 3.62**Rounding Off a Number in Decimal Form to Three Digits**

Round 5.67849 to three decimal digits.

 Solution

The third decimal digit is 8. The digit following the 8 is 4. When the digit is 4, we write the number only to the third digit. So, 5.67849 rounded off to three decimal places is 5.678.

 YOUR TURN 3.62

1. Round 5.1082 to three decimal places.

EXAMPLE 3.63**Rounding Off a Number in Decimal Form to Four Digits**

Round 45.11475 to four decimal digits.

✓ **Solution**

The fourth decimal digit is 7. The digit following the 7 is 5. When the digit is 5, we write the number only to the fourth decimal digit, 45.1147. We then replace the fourth decimal digit by one more than the fourth digit, which yields 45.1148. So, 45.11475 rounded off to four decimal places is 45.1148.

➤ **YOUR TURN 3.63**

1. Round 18.6298 to two decimal places.

To convert a rational number in fraction form to decimal form, use your calculator to perform the division.

EXAMPLE 3.64**Converting a Rational Number in Fraction Form into Decimal Form**

Convert $\frac{47}{25}$ into decimal form.

✓ **Solution**

Using a calculator to divide 47 by 25, the result is 1.88.

➤ **YOUR TURN 3.64**

1. Convert $\frac{48}{30}$ into decimal form.

Converting a terminating decimal to the fractional form may be done in the following way:

Step 1: Count the number of digits in the decimal part of the number, labeled n .

Step 2: Raise 10 to the n th power.

Step 3: Rewrite the number without the decimal.

Step 4: The fractional form is the number from Step 3 divided by the result from Step 2.

This process works due to what decimals represent and how we work with mixed numbers. For example, we could convert the number 7.4536 to fractional form. The decimal part of the number, the .4536 part of 7.4536, has four digits. By the definition of decimal notation, the decimal portion represents $\frac{4,536}{10^4} = \frac{4,536}{10,000}$. The decimal number 7.4536 is equal to the improper fraction $7\frac{4,536}{10,000}$. Adding those to fractions yields $\frac{74,536}{10,000}$.

EXAMPLE 3.65**Converting from Decimal Form to Fraction Form with Terminating Decimals**

Convert 3.2117 to fraction form.

✓ **Solution**

Step 1: There are four digits after the decimal point, so $n = 4$.

Step 2: Raise 10 to the fourth power, $10^4 = 10,000$.

Step 3: When we remove the decimal point, we have 32,117.

Step 4: The fraction has as its numerator the result from Step 3 and as its denominator the result of Step 2, which is the

fraction $\frac{32,117}{10,000}$.

YOUR TURN 3.65

1. Convert 17.03347 to fraction form.

The process is different when converting from the decimal form of a rational number into fraction form when the decimal form is a repeating decimal. This process is not covered in this text.

Multiplying and Dividing Rational Numbers

Multiplying rational numbers is less complicated than adding or subtracting rational numbers, as there is no need to find common denominators. To multiply rational numbers, multiply the numerators, then multiply the denominators, and write the numerator product divided by the denominator product. Symbolically, $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$. As always, rational numbers should be reduced to lowest terms.

FORMULA

If b and d are non-zero integers, then $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$.

EXAMPLE 3.66

Multiplying Rational Numbers

Calculate $\frac{12}{25} \times \frac{10}{21}$.

Solution

Multiply the numerators and place that in the numerator, and then multiply the denominators and place that in the denominator.

$$\frac{12}{25} \times \frac{10}{21} = \frac{12 \times 10}{25 \times 21} = \frac{120}{525}$$

This is not in lowest terms, so this needs to be reduced. The GCD of 120 and 525 is 15.

$$\frac{120}{525} = \frac{15 \times 8}{15 \times 35} = \frac{8}{35}$$

YOUR TURN 3.66

1. Calculate $\frac{45}{88} \times \frac{28}{75}$.

VIDEO

[Multiplying Fractions \(https://openstax.org/r/Multiplying_Fractions\)](https://openstax.org/r/Multiplying_Fractions)

As with multiplication, division of rational numbers can be done using a calculator.

EXAMPLE 3.67

Dividing Decimals with a Calculator

Calculate $3.45 \div 2.341$ using a calculator. Round to three decimal places if necessary.

Solution

Using a calculator, we obtain 1.473729175565997. Rounding to three decimal places we have 1.474.

> YOUR TURN 3.67

1. Calculate $45.63 \div 17.13$ using a calculator. Round to three decimal places, if necessary.

Before discussing division of fractions without a calculator, we should look at the reciprocal of a number. The **reciprocal** of a number is 1 divided by the number. For a fraction, the reciprocal is the fraction formed by switching the numerator and denominator. For the fraction $\frac{a}{b}$, the reciprocal is $\frac{b}{a}$. An important feature for a number and its reciprocal is that their product is 1.

When dividing two fractions by hand, find the reciprocal of the **divisor** (the number that is being divided into the other number). Next, replace the divisor by its reciprocal and change the division into multiplication. Then, perform the multiplication. Symbolically, $\frac{b}{a} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$. As before, reduce to lowest terms.

FORMULA

If b, c and d are non-zero integers, then $\frac{b}{a} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$.

EXAMPLE 3.68**Dividing Rational Numbers**

1. Calculate $\frac{4}{21} \div \frac{6}{35}$.
2. Calculate $\frac{1}{8} \div \frac{5}{28}$.

✓ Solution

1. **Step 1:** Find the reciprocal of the number being divided by $\frac{6}{35}$. The reciprocal of that is $\frac{35}{6}$.

Step 2: Multiply the first fraction by that reciprocal.

$$\frac{4}{21} \div \frac{6}{35} = \frac{4}{21} \times \frac{35}{6} = \frac{140}{126}$$

The answer, $\frac{140}{126}$ is not in lowest terms. The GCD of 140 and 126 is 14. Factoring and canceling gives

$$\frac{140}{126} = \frac{14 \times 10}{14 \times 9} = \frac{10}{9}.$$

2. **Step 1:** Find the reciprocal of the number being divided by, which is $\frac{5}{28}$. The reciprocal of that is $\frac{28}{5}$.

Step 2: Multiply the first fraction by that reciprocal: $\frac{1}{8} \div \frac{5}{28} = \frac{1}{8} \times \frac{28}{5} = \frac{28}{40}$

The answer, $\frac{28}{40}$, is not in lowest reduced form. The GCD of 28 and 40 is 4. Factoring and canceling gives

$$\frac{28}{40} = \frac{4 \times 7}{4 \times 10} = \frac{7}{10}.$$

> YOUR TURN 3.68

1. Calculate $\frac{46}{175} \div \frac{69}{285}$.
2. Calculate $\frac{3}{40} \div \frac{42}{55}$.

▶ VIDEO

[Dividing Fractions \(https://openstax.org/r/Dividing_Fractions\)](https://openstax.org/r/Dividing_Fractions)

Applying the Order of Operations to Simplify Expressions

The order of operations for rational numbers is the same as for integers, as discussed in [Order of Operations](#). The order of operations makes it easier for anyone to correctly calculate and represent. The order follows the well-known acronym PEMDAS:

P	Parentheses
E	Exponents
M/D	Multiplication and division
A/S	Addition and subtraction

The first step in calculating using the order of operations is to perform operations inside the parentheses. Moving down the list, next perform all exponent operations moving from left to right. Next (left to right once more), perform all multiplications and divisions. Finally, perform the additions and subtractions.

EXAMPLE 3.69

Applying the Order of Operations with Rational Numbers

Correctly apply the rules for the order of operations to accurately compute $(\frac{5}{7} - \frac{2}{7}) \times 2^3$.

Solution

Step 1: To calculate this, perform all calculations within the parentheses before other operations.

$$(\frac{5}{7} - \frac{2}{7}) \times 2^3 = (\frac{3}{7}) \times 2^3$$

Step 2: Since all parentheses have been cleared, we move left to right, and compute all the exponents next.

$$(\frac{3}{7}) \times 2^3 = (\frac{3}{7}) \times 8$$

Step 3: Now, perform all multiplications and divisions, moving left to right.

$$(\frac{3}{7}) \times 8 = \frac{24}{7}$$

YOUR TURN 3.69

1. Correctly apply the rules for the order of operations to accurately compute $(\frac{3}{16} + \frac{7}{16})^2 + \frac{1}{5} \div \frac{3}{10}$.

EXAMPLE 3.70

Applying the Order of Operations with Rational Numbers

Correctly apply the rules for the order of operations to accurately compute $4 + \frac{2}{3} \div \left(\left(\frac{5}{9} \right)^2 - \left(\frac{2}{3} + 5 \right) \right)^2$.

Solution

To calculate this, perform all calculations within the parentheses before other operations. Evaluate the innermost parentheses first. We can work separate parentheses expressions at the same time.

Step 1: The innermost parentheses contain $\frac{2}{3} + 5$. Calculate that first, dividing after finding the common denominator.

$$\begin{aligned} & 4 + \frac{2}{3} \div \left(\left(\frac{5}{9} \right)^2 - \left(\frac{2}{3} + 5 \right) \right)^2 \\ &= 4 + \frac{2}{3} \div \left(\left(\frac{5}{9} \right)^2 - \left(\frac{2}{3} + \frac{5}{1} \right) \right)^2 \\ &= 4 + \frac{2}{3} \div \left(\left(\frac{5}{9} \right)^2 - \left(\frac{2}{3} + \frac{15}{3} \right) \right)^2 \\ &= 4 + \frac{2}{3} \div \left(\left(\frac{5}{9} \right)^2 - \left(\frac{17}{3} \right) \right)^2 \end{aligned}$$

Step 2: Calculate the exponent in the parentheses, $\left(\frac{5}{9} \right)^2$.

$$4 + \frac{2}{3} \div \left(\left(\frac{5}{9} \right)^2 - \left(\frac{17}{3} \right) \right)^2$$

$$= 4 + \frac{2}{3} \div \left(\left(\frac{25}{81} \right) - \left(\frac{17}{3} \right) \right)^2$$

Step 3: Subtract inside the parentheses is done, using a common denominator.

$$4 + \frac{2}{3} \div \left(\left(\frac{25}{81} \right) - \left(\frac{17}{3} \right) \right)^2$$

$$4 + \frac{2}{3} \div \left(\left(\frac{25}{81} \right) - \left(\frac{17 \times 27}{3 \times 27} \right) \right)^2$$

$$4 + \frac{2}{3} \div \left(\left(\frac{25}{81} \right) - \left(\frac{459}{81} \right) \right)^2$$

$$4 + \frac{2}{3} \div \left(\left(\frac{-434}{81} \right) \right)^2$$

Step 4: At this point, evaluate the exponent and divide.

$$4 + \frac{2}{3} \div \left(\left(\frac{-434}{81} \right) \right)^2$$

$$4 + \frac{2}{3} \div \left(\frac{188,356}{6,561} \right)$$

$$= 4 + \frac{2}{3} \times \left(\frac{6,561}{188,356} \right)$$

$$= 4 + \frac{2,187}{94,178}$$

Step 5: Add.

$$4 + \frac{2,187}{94,178}$$

$$= \frac{378,899}{94,178}$$

Had this been done on a calculator, the decimal form of the answer would be 4.0232 (rounded to four decimal places).

> YOUR TURN 3.70

1. Correctly apply the rules for the order of operations to accurately compute $\left(\frac{3}{5} + 2\right) \times \left(\frac{4}{5} - \frac{1}{2}\right)^2 \div \frac{11}{15}$.

▶ VIDEO

[Order of Operations Using Fractions \(https://openstax.org/r/Operations_Using_Fractions\)](https://openstax.org/r/Operations_Using_Fractions)

Applying the Density Property of Rational Numbers

Between any two rational numbers, there is another rational number. This is called the **density property** of the rational numbers.

Finding a rational number between any two rational numbers is very straightforward.

Step 1: Add the two rational numbers.

Step 2: Divide that result by 2.

The result is always a rational number. This follows what we know about rational numbers. If two fractions are added, then the result is a fraction. Also, when a fraction is divided by a fraction (and 2 is a fraction), then we get another fraction. This two-step process will give a rational number, provided the first two numbers were rational.

EXAMPLE 3.71

Applying the Density Property of Rational Numbers

Demonstrate the density property of rational numbers by finding a rational number between $\frac{4}{11}$ and $\frac{7}{12}$.

 Solution

To find a rational number between $\frac{4}{11}$ and $\frac{7}{12}$:

Step 1: Add the fractions.

$$\frac{4}{11} + \frac{7}{12} = \frac{4 \times 12}{11 \times 12} + \frac{7 \times 11}{12 \times 11} = \frac{48}{132} + \frac{77}{132} = \frac{125}{132}$$

Step 2: Divide the result by 2. Recall that to divide by 2, you multiply by the reciprocal of 2. The reciprocal of 2 is $\frac{1}{2}$, as seen below.

$$\frac{125}{132} \div 2 = \frac{125}{132} \times \frac{1}{2} = \frac{125}{264}$$

So, one rational number between $\frac{4}{11}$ and $\frac{7}{12}$ is $\frac{125}{264}$.

We could check that the number we found is between the other two by finding the decimal representation of the numbers. Using a calculator, the decimal representations of the rational numbers are 0.363636..., 0.473484848..., and 0.5833333.... Here it is clear that $\frac{125}{264}$ is between $\frac{4}{11}$ and $\frac{7}{12}$.

 YOUR TURN 3.71

1. Demonstrate the density property of rational numbers by finding a rational number between $\frac{27}{13}$ and $\frac{21}{10}$.

Solving Problems Involving Rational Numbers

Rational numbers are used in many situations, sometimes to express a portion of a whole, other times as an expression of a ratio between two quantities. For the sciences, converting between units is done using rational numbers, as when converting between gallons and cubic inches. In chemistry, mixing a solution with a given concentration of a chemical per unit volume can be solved with rational numbers. In demographics, rational numbers are used to describe the distribution of the population. In dietetics, rational numbers are used to express the appropriate amount of a given ingredient to include in a recipe. As discussed, the application of rational numbers crosses many disciplines.

EXAMPLE 3.72**Mixing Soil for Vegetables**

James is mixing soil for a raised garden, in which he plans to grow a variety of vegetables. For the soil to be suitable, he determines that $\frac{2}{5}$ of the soil can be topsoil, but $\frac{2}{5}$ needs to be peat moss and $\frac{1}{5}$ has to be compost. To fill the raised garden bed with 60 cubic feet of soil, how much of each component does James need to use?

 Solution

In this example, we know the proportion of each component to mix, and we know the total amount of the mix we need. In this kind of situation, we need to determine the appropriate amount of each component to include in the mixture. For each component of the mixture, multiply 60 cubic feet, which is the total volume of the mixture we want, by the fraction required of the component.

Step 1: The required fraction of topsoil is $\frac{2}{5}$, so James needs $60 \times \frac{2}{5}$ cubic feet of topsoil. Performing the multiplication, James needs $60 \times \frac{2}{5} = \frac{120}{5} = 24$ (found by treating the fraction as division, and 120 divided by 5 is 24) cubic feet of topsoil.

Step 2: The required fraction of peat moss is also $\frac{2}{5}$, so he also needs $60 \times \frac{2}{5}$ cubic feet, or $60 \times \frac{2}{5} = \frac{120}{5} = 24$ cubic feet of peat moss.

Step 3: The required fraction of compost is $\frac{1}{5}$. For the compost, he needs $60 \times \frac{1}{5} = \frac{60}{5} = 12$ cubic feet.

 YOUR TURN 3.72

1. Ashley wants to study for 10 hours over the weekend. She plans to spend half the time studying math, a quarter

of the time studying history, an eighth of the time studying writing, and the remaining eighth of the time studying physics. How much time will Ashley spend on each of those subjects?

EXAMPLE 3.73

Determining the Number of Specialty Pizzas

At Bella's Pizza, one-third of the pizzas that are ordered are one of their specialty varieties. If there are 273 pizzas ordered, how many were specialty pizzas?

Solution

One-third of the whole are specialty pizzas, so we need one-third of 273, which gives $\frac{1}{3} \times 273 = \frac{273}{3} = 91$, found by dividing 273 by 3. So, 91 of the pizzas that were ordered were specialty pizzas.

YOUR TURN 3.73

1. Danny, a nutritionist, is designing a diet for her client, Callum. Danny determines that Callum's diet should be 30% protein. If Callum consumes 2,400 calories per day, how many calories of protein should Danny tell Callum to consume?

VIDEO

Finding a Fraction of a Total (https://openstax.org/r/Finding_Fraction_of_Total)

Using Fractions to Convert Between Units

A common application of fractions is called **unit conversion**, or **converting units**, which is the process of changing from the units used in making a measurement to different units of measurement.

For instance, 1 inch is (approximately) equal to 2.54 cm. To convert between units, the two equivalent values are made into a fraction. To convert from the first type of unit to the second type, the fraction has the second unit as the numerator, and the first unit as the denominator.

From the inches and centimeters example, to change from inches to centimeters, we use the fraction $\frac{2.54 \text{ cm}}{1 \text{ in}}$. If, on the other hand, we wanted to convert from centimeters to inches, we'd use the fraction $\frac{1 \text{ in}}{2.54 \text{ cm}}$. This fraction is multiplied by the number of units of the type you are converting *from*, which means the units of the denominator are the same as the units being multiplied.

EXAMPLE 3.74

Converting Liters to Gallons

It is known that 1 liter (L) is 0.264172 gallons (gal). Use this to convert 14 liters into gallons.

Solution

We know that 1 liter = 0.264172 gal. Since we are converting from liters, when we create the fraction we use, make sure the liter part of the equivalence is in the denominator. So, to convert the 14 liters to gallons, we multiply 14 by $\frac{1 \text{ gal}}{0.264172 \text{ gal / 1 liter}}$. Notice the gallon part is in the numerator since we're converting *to* gallons, and the liter part is in the denominator since we are converting *from* liters. Performing this and rounding to three decimal places, we find that 14 liters is $14 \text{ liter} \times \frac{0.264172 \text{ gal}}{1 \text{ liter}} = 3.69841 \text{ gal}$.

YOUR TURN 3.74

1. One mile is equal to 1.60934 km. Convert 200 miles to kilometers. Round off the answer to three decimal places.

EXAMPLE 3.75**Converting Centimeters to Inches**

It is known that 1 inch is 2.54 centimeters. Use this to convert 100 centimeters into inches.

 Solution

We know that 1 inch = 2.54 cm. Since we are converting from centimeters, when we create the fraction we use, make sure the centimeter part of the equivalence is in the denominator, $\frac{1 \text{ in}}{2.54 \text{ cm}}$. To convert the 100 cm to inches, multiply 100 by $\frac{1 \text{ in}}{2.54 \text{ cm}}$. Notice the inch part is in the numerator since we're converting *to* inches, and the centimeter part is in the denominator since we are converting *from* centimeters. Performing this and rounding to three decimal places, we obtain $100 \text{ cm} \times \frac{1 \text{ in}}{2.54 \text{ cm}} = 39.370 \text{ in}$. This means 100 cm equals 39.370 in.

 YOUR TURN 3.75

1. It is known that 4 quarts equals 3.785 liters. If you have 25 quarts, how many liters do you have? Round off to three decimal places.

 VIDEO

[Converting Units \(https://openstax.org/r/Converting_Units\)](https://openstax.org/r/Converting_Units)

Defining and Applying Percent

A **percent** is a specific rational number and is literally per 100. n percent, denoted $n\%$, is the fraction $\frac{n}{100}$.

EXAMPLE 3.76**Rewriting a Percentage as a Fraction**

Rewrite the following as fractions:

1. 31%
2. 93%

 Solution

1. Using the definition and $n = 31$, 31% in fraction form is $\frac{31}{100}$.
2. Using the definition and $n = 93$, 93% in fraction form is $\frac{93}{100}$.

 YOUR TURN 3.76

Rewrite the following as fractions:

1. 4%
2. 50%

EXAMPLE 3.77**Rewriting a Percentage as a Decimal**

Rewrite the following percentages in decimal form:

1. 54%
2. 83%

 Solution

1. Using the definition and $n = 54$, 54% in fraction form is $\frac{54}{100}$. Dividing a number by 100 moves the decimal two

- places to the left; 54% in decimal form is then 0.54.
- Using the definition and $n = 83$, 83% in fraction form is $\frac{83}{100}$. Dividing a number by 100 moves the decimal two places to the left; 83% in decimal form is then 0.83.

YOUR TURN 3.77

Rewrite the following percentages in decimal form:

- 14%
- 7%

You should notice that you can simply move the decimal two places to the left without using the fractional definition of percent.

Percent is used to indicate a fraction of a total. If we want to find 30% of 90, we would perform a multiplication, with 30% written in either decimal form or fractional form. The 90 is the **total**, 30 is the **percentage**, and 27 (which is 0.30×90) is the **percentage of the total**.

FORMULA

$n\%$ of x items is $\frac{n}{100} \times x$. The x is referred to as the **total**, the n is referred to as the **percent** or **percentage**, and the value obtained from $\frac{n}{100} \times x$ is the **part** of the total and is also referred to as the **percentage of the total**.

EXAMPLE 3.78

Finding a Percentage of a Total

- Determine 40% of 300.
- Determine 64% of 190.

Solution

- The total is 300, and the percentage is 40. Using the decimal form of 40% and multiplying we obtain $0.40 \times 300 = 120$.
- The total is 190, and the percentage is 64. Using the decimal form of 64% and multiplying we obtain $0.64 \times 190 = 121.6$.

YOUR TURN 3.78

- Determine 25% of 1,200.
- Determine 53% of 1,588.

In the previous situation, we knew the total and we found the percentage of the total. It may be that we know the percentage of the total, and we know the percent, but we don't know the total. To find the total if we know the percentage of the total, use the following formula.

FORMULA

If we know that $n\%$ of the total is x , then the total is $\frac{100 \times x}{n}$.

EXAMPLE 3.79**Finding the Total When the Percentage and Percentage of the Total Are Known**

1. What is the total if 28% of the total is 140?
2. What is the total if 6% of the total is 91?

✓ Solution

1. 28 is the percentage, so $n = 28$. 28% of the total is 140, so $x = 140$. Using those we find that the total was $\frac{100 \times 140}{28} = 500$.
2. 6 is the percentage, so $n = 6$. 6% of the total is 91, so $x = 91$. Using those we find that the total was $\frac{100 \times 91}{6} = 1,516.6$.

> YOUR TURN 3.79

1. What is the total if 25% of the total is 30?
2. What is the total if 45% of the total is 360?

The percentage can be found if the total and the percentage of the total is known. If you know the total, and the percentage of the total, first divide the part by the total. Move the decimal two places to the right and append the symbol %. The percentage may be found using the following formula.

FORMULA

The percentage, n , of b that is a is $\frac{a}{b} \times 100\%$.

EXAMPLE 3.80**Finding the Percentage When the Total and Percentage of the Total Are Known**

Find the percentage in the following:

1. Total is 300, percentage of the total is 60.
2. Total is 440, percentage of the total is 176.

✓ Solution

1. The total is 300; the percentage of the total is 60. Calculating yields 0.2. Moving the decimal two places to the right gives 20. Appending the percentage to this number results in 20%. So, 60 is 20% of 300.
2. The total is 440; the percentage of the total is 176. Calculating yields 0.4. Moving the decimal two places to the right gives 40. Appending the percentage to this number results in 40%. So, 176 is 40% of 440.

> YOUR TURN 3.80

Find the percentage in the following:

1. Total is 1,000, percentage of the total is 70.
2. Total is 500, percentage of the total is 425.

Solve Problems Using Percent

In the media, in research, and in casual conversation percentages are used frequently to express proportions. Understanding how to use percent is vital to consuming media and understanding numbers. Solving problems using percentages comes down to identifying which of the three components of a percentage you are given, the total, the percentage, or the percentage of the total. If you have two of those components, you can find the third using the methods outlined previously.

EXAMPLE 3.81**Percentage of Students Who Are Sleep Deprived**

A study revealed that 70% of students suffer from sleep deprivation, defined to be sleeping less than 8 hours per night. If the survey had 400 participants, how many of those participants had less than 8 hours of sleep per night?

 Solution

The percentage of interest is 70%. The total number of students is 400. With that, we can find how many were in the percentage of the total, or, how many were sleep deprived. Applying the formula from above, the number who were sleep deprived was $0.70 \times 400 = 280$; 280 students on the study were sleep deprived.

 YOUR TURN 3.81

1. Riley has a daily calorie intake of 2,200 calories and wants to take in 20% of their calories as protein. How many calories of protein should be in their daily diet?

EXAMPLE 3.82**Amazon Prime Subscribers**

There are 126 million users who are U.S. Amazon Prime subscribers. If there are 328.2 million residents in the United States, what percentage of U.S. residents are Amazon Prime subscribers?

 Solution

We are asked to find the percentage. To do so, we divide the percentage of the total, which is 126 million, by the total, which is 328.2 million. Performing this division and rounding to three decimal places yields $\frac{126}{328.2} = 0.384$. The decimal is moved to the right by two places, and a % sign is appended to the end. Doing this shows us that 38.4% of U.S. residents are Amazon Prime subscribers.

 YOUR TURN 3.82

1. A small town has 450 registered voters. In the primaries, 54 voted. What percentage of registered voters in that town voted in the primaries?

EXAMPLE 3.83**Finding the Percentage When the Total and Percentage of the Total Are Known**

Evander plays on the basketball team at their university and 73% of the athletes at their university receive some sort of scholarship for attending. If they know 219 of the student-athletes receive some sort of scholarship, how many student-athletes are at the university?

 Solution

We need to find the total number of student-athletes at Evander's university.

Step 1: Identify what we know. We know the percentage of students who receive some sort of scholarship, 73%. We also know the number of athletes that form the part of the whole, or 219 student-athletes.

Step 2: To find the total number of student-athletes, use $\frac{100 \times x}{n}$, with $x = 219$ and $n = 73$. Calculating with those values yields $\frac{100 \times 219}{73} = 300$.

So, there are 300 total student-athletes at Evander's university

 YOUR TURN 3.83

1. A store declares a deep discount of 40% for an item, which they say will save \$30. What was the original price of the item?

Check Your Understanding

17. Identify which of the following are rational numbers.
 -41 , $\sqrt{13}$, $\frac{4}{3}$, 2.75 , $0.2\overline{13}$
18. Express $\frac{18}{30}$ in lowest terms.
19. Calculate $\frac{3}{8} + \frac{5}{12}$ and express in lowest terms.
20. Convert 0.34 into fraction form.
21. Convert $\frac{47}{12}$ into a mixed number.
22. Calculate $\frac{2}{9} \times \frac{21}{22}$ and express in lowest terms.
23. Calculate $\frac{2}{5} \div \frac{3}{10} + \frac{1}{6}$.
24. Identify a rational number between $\frac{7}{8}$ and $\frac{20}{21}$.
25. Rewrite $3\frac{2}{7}$ as an improper fraction.
26. Lina decides to save $\frac{1}{8}$ of her take-home pay every paycheck. Her most recent paycheck was for \$882. How much will she save from that paycheck?
27. Determine 38% of 600.
28. A microchip factory has decided to increase its workforce by 10%. If it currently has 70 employees, how many new employees will the factory hire?



SECTION 3.4 EXERCISES

For the following exercises, identify which of the following are rational numbers.

1. 4.598
2. $\sqrt{144}$
3. $\sqrt{131}$

For the following exercises, reduce the fraction to lowest terms

4. $\frac{8}{10}$
5. $\frac{30}{105}$
6. $\frac{36}{539}$
7. $\frac{231}{490}$
8. $\frac{750}{17,875}$

For the following exercises, do the indicated conversion. If it is a repeating decimal, use the correct notation.

9. Convert $\frac{25}{6}$ to a mixed number.
10. Convert $\frac{240}{53}$ to a mixed number.
11. Convert $2\frac{3}{8}$ to an improper fraction.
12. Convert $15\frac{7}{30}$ to an improper fraction.
13. Convert $\frac{4}{9}$ to decimal form.
14. Convert $\frac{13}{20}$ to decimal form.
15. Convert $\frac{27}{625}$ to decimal form.
16. Convert $\frac{11}{14}$ to decimal form.
17. Convert 0.23 to fraction form and reduce to lowest terms.

18. Convert 3.8874 to fraction form and reduce to lowest terms.

For the following exercises, perform the indicated operations. Reduce to lowest terms.

19. $\frac{3}{5} + \frac{3}{10}$
20. $\frac{3}{14} + \frac{8}{21}$
21. $\frac{13}{36} - \frac{14}{99}$
22. $\frac{13}{24} - \frac{4}{117}$
23. $\frac{3}{7} \times \frac{21}{48}$
24. $\frac{48}{143} \times \frac{77}{120}$
25. $\frac{14}{27} \div \frac{7}{12}$
26. $\frac{44}{75} \div \frac{484}{285}$
27. $\left(\frac{3}{5} + \frac{2}{7}\right) \times \frac{10}{21}$
28. $\frac{3}{8} \times \left(\frac{13}{12} - \frac{35}{36}\right)$
29. $\left(\frac{3}{7} + \frac{5}{16}\right)^2 - \frac{5}{12}$
30. $\frac{3}{8} \times \left(\frac{4}{9} - \frac{1}{8}\right)^2$
31. $\left(\frac{2}{5} \times \left(\frac{7}{8} - \frac{2}{3}\right)\right)^2 \div \left(\frac{4}{9} + \frac{5}{6}\right) + \frac{7}{12}$
32. $\left(\frac{1}{5} \div \left(\frac{3}{10} + \frac{11}{15}\right)\right) \times \left(\frac{2}{21} + \frac{5}{9}\right) - \left(\frac{8}{15} \div \frac{4}{33}\right)^2$
33. Find a rational number between $\frac{8}{17}$ and $\frac{15}{28}$
34. Find a rational number between $\frac{3}{50}$ and $\frac{13}{98}$.
35. Find two rational numbers between $\frac{3}{10}$ and $\frac{19}{45}$.
36. Find three rational numbers between $\frac{5}{12}$ and $\frac{175}{308}$.
37. Convert 24% to fraction form and reduce completely.
38. Convert 95% to fraction form and reduce completely.
39. Convert 0.23 to a percentage.
40. Convert 1.22 to a percentage.
41. Determine 30% of 250.
42. Determine 75% of 600.
43. If 25% of a group is 41 members, how many members total are in the group?
44. If 80% of the total is 60, how much is in the total?
45. 13 is what percent of 20?
46. 80 is what percent of 320?
47. Professor Donalson's history of film class has 60 students. Of those students, $\frac{2}{5}$ say their favorite movie genre is comedy. How many of the students in Professor Donalson's class name comedy as their favorite movie genre?
48. Naia's dormitory floor has 80 residents. Of those, $\frac{3}{8}$ play *Fortnight* for at least 15 hours per week. How many students on Naia's floor play *Fortnight* at least 15 hours per week?
49. In Tara's town there are 24,000 people. Of those, $\frac{13}{100}$ are food insecure. How many people in Tara's town are food insecure?
50. Roughly $\frac{4}{5}$ of air is nitrogen. If an enclosure holds 2,000 liters of air, how many liters of nitrogen should be expected in the enclosure?
51. To make the dressing for coleslaw, Maddie needs to mix it with $\frac{3}{5}$ mayonnaise and $\frac{2}{5}$ apple cider vinegar. If Maddie wants to have 8 cups of dressing, how many cups of mayonnaise and how many cups of apple cider vinegar does Maddie need?
52. Malika is figuring out their schedule. They wish to spend $\frac{4}{15}$ of their time sleeping, $\frac{1}{3}$ of their time studying and going to class, $\frac{1}{5}$ of their time at work, and $\frac{2}{15}$ of their time doing other activities, such as entertainment or exercising. There are 168 hours in a week. How many hours in a week will Malika spend:
 - a. Sleeping?
 - b. Studying and going to class?
 - c. Not sleeping?
53. Roughly 20.9% of air is oxygen. How much oxygen is there in 200 liters of air?
54. 65% of college students graduate within 6 years of beginning college. A first-year cohort at a college contains 400 students. How many are expected to graduate within 6 years?

55. A 20% discount is offered on a new laptop. How much is the discount if the new laptop originally cost \$700?
56. Leya helped at a neighborhood sale and was paid 5% of the proceeds. If Leya is paid \$171.25, what were the total proceeds from the neighborhood sale?
57. **Unit Conversion.** 1 kilogram (kg) is equal to 2.20462 pounds. Convert 13 kg to pounds. Round to three decimal places, if necessary.
58. **Unit Conversion.** 1 kilogram (kg) is equal to 2.20462 pounds. Convert 200 pounds to kilograms. Round to three decimal places, if necessary.
59. **Unit Conversion.** There are 12 inches in a foot, 3 feet in a yard, and 1,760 yards in a mile. Convert 10 miles to inches. To do so, first convert miles to yards. Next, convert the yards to feet. Last, convert the feet to inches.
60. **Unit Conversion.** There are 1,000 meters (m) in a kilometer (km), and 100 centimeters (cm) in a meter. Convert 4 km to centimeters.
61. **Markup.** In this exercise, we introduce the concept of **markup**. The **markup** on an item is the difference between how much a store sells an item for and how much the store paid for the item. Suppose Wegmans (a northeastern U.S. grocery chain) buys cereal at \$1.50 per box and sells the cereal for \$2.29.
- Determine the markup in dollars.
 - The markup is what percent of the original cost? Round the percentage to one decimal place.
62. In this exercise, we explore what happens when an item is marked up by a percentage, and then marked down using the same percentage. Wegmans purchases an item for \$5 per unit. The markup on the item is 25%.
- Calculate the markup on the item, in dollars.
 - What is the price for which Wegmans sells the item? This is the price Wegmans paid, plus the markup.
 - Suppose Wegmans then offers a 25% discount on the sale price of the item (found in part b). In dollars, how much is the discount?
 - Determine the price of the item after the discount (this is the sales price of the item minus the discount). Round to two decimal places.
 - Is the new price after the markup and discount equal to the price Wegmans paid for the item? Explain.
63. **Repeated Discounts.** In this exercise, we explore applying more than one discount to an item. Suppose a store cuts the price on an item by 50%, and then offers a coupon for 25% off any sale item. We will find the price of the item after applying the sale price and the coupon discount.
- The original price was \$150. After the 50% discount, what is the price of the item?
 - The coupon is applied to the discount price. The coupon is for 25%. Find 25% of the sale price (found in part a).
 - Find the price after applying the coupon (this is the value from part a minus the value from part b).
 - The total amount saved on the item is the original price after all the discounts. Determine the total amount saved by subtracting the final price paid (part c) from the original price of the item.
 - Determine the effective discount percentage, which is the total amount saved divided by the original price of the item.
 - Was the effective discount percentage equal to 75%, which would be the 50% plus the 25%? Explain.

Converting Repeating Decimals to a Fraction

It was mentioned in the section that repeating decimals are rational numbers. To convert a repeating decimal to a rational number, perform the following steps:

Step 1: Label the original number S .

Step 2: Count the number of digits, n , in the repeating part of the number.

Step 3: Multiply S by 10^n , and label this as $10^n \times S$.

Step 4: Determine $10^n - 1$.

Step 5: Calculate $10^n \times S - S$. If done correctly, the repeating part of the number will cancel out.

Step 6: If the result from Step 5 has decimal digits, count the number of decimal digits in the number from Step 5. Label this m .

Step 7: Remove the decimal from the result of Step 5.

Step 8: Add m zeros to the end of the number from Step 4.

Step 9: Divide the result from Step 7 by the result from Step 8. This is the fraction form of the repeating decimal.

- Convert $0.\overline{7}$ to fraction form.
- Convert $0.\overline{45}$ to fraction form.
- Convert $3.1\overline{5}$ to fraction form.
- Convert $2.71\overline{94}$ to fraction form.